

VHGRAD

Development of a Didactic Deep Learning Framework
with a Custom Autograd Engine

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Technical Progression of the VHgrad Framework

1. Foundations of Neural Networks

Introduction to neural network concepts, and core components such as neurons, layers and parameters.

2. Initial Implementations

Manual implementation of single neurons and multi-neuron layers with emphasis on mathematical foundations.

3. Vectorized Computation

Transition to NumPy-based implementations using vectors, matrices, batching, and efficient matrix operations.

4. Multi-Layer Network Construction

Encapsulation of dense layers into reusable classes and structured handling of training data.

5. Activation Functions

Implementation of linear and non-linear activation functions, including Sigmoid, ReLU, and Softmax.

6. Loss Functions and Evaluation

Integration of categorical cross-entropy loss and accuracy metrics for model evaluation.

7. Optimization and Backpropagation

Gradient-based optimization, derivatives, chain rule, and full backpropagation pipeline.

8. Optimizers

Implementation of SGD, learning rate decay, momentum, AdaGrad, RMSProp, and Adam.

9. Regularization Techniques

L1 and L2 regularization, dropout, and validation strategies to improve generalization.

10. Classification and Regression

Support for binary classification and regression tasks with appropriate loss functions.

11. Model Abstraction

Introduction of a unified model object encapsulating layers, losses, and optimizers.

12. Real-World Data Integration

Data loading, preprocessing, shuffling, batching, and training on real datasets.

13. Model Persistence and Inference

Saving and loading model parameters and enabling prediction on unseen data.